

BVC24 ERP

AI SMART MANUFACTURING

Employee Module Documentation

Deployment & User Guide

Bharath Vending Corporation

Vendor-based Manufacturing ERP

Version: 1.0

Date: 6 June 2026

Audience: HR | IT | Employees | Implementers

BVC24 ERP - Complete System Documentation

Product: Bharath Vending Corporation - Vendor-based Manufacturing ERP **Prepared for:** Bharath Vending Corporation **Version:** 1.0 **Documentation date:** 2026-07-10

1. Executive Summary

BVC24 ERP is an integrated, on-premise Enterprise Resource Planning system built specifically for **Bharath Vending Corporation**, a vending-machine manufacturer serving Tier-1 and Tier-2 markets across India. The platform consolidates every business-critical workflow - from lead capture through manufacturing, dispatch, payroll and after-sales - into a single browser-accessible application backed by an in-office server.

The system serves three primary user groups:

- **Employees** - self-service portal for attendance, leave, memos, payslips and task updates
- **Managers / HR** - team dashboards, approvals, reports, and performance analytics
- **Administrators** - full control over master data, users, RBAC, deployment, and system settings

Key differentiators of this implementation:

- **Biometric-driven attendance** integrated live with an eSSL X2008 fingerprint device
- **AI-powered HR assistant** using self-hosted LLMs (Ollama with Qwen 2.5), with a voice-first roadmap
- **Zero third-party dependencies** for AI or attendance - everything runs on the office server
- **Real-time payroll calculation** driven by biometric attendance plus configurable salary structures
- **Multi-tenant ready** via a `VENDOR_ID` scope on every business table

2. Business Objectives

BVC24 ERP was commissioned to address the following operational pain points:

1. **Manual attendance recording:** Register-based attendance was error-prone and disconnected from payroll.
2. **Fragmented data:** Employee, sales, project, and inventory data lived in separate spreadsheets.
3. **Slow decision-making:** HR and management had no live view of workforce productivity, project progress, or cash position.
4. **Payroll cycle burden:** End-of-month salary calculation required 3-4 days of manual reconciliation.
5. **Compliance risk:** Employee data, salary structures, and PF/ESI records were stored on individual laptops with no audit trail.

Business benefits realised:

Metric	Before	After BVC24 ERP
Attendance capture latency	End of day (paper)	Live via biometric bridge (< 2 min)
Payroll cycle time	3-4 days	Same-day (auto-computed)
Employee self-service queries to HR	~40 / week	~5 / week
Task assignment	Verbal / whiteboard	Structured, tracked, reportable
Onboarding paperwork	2 days per hire	20 minutes

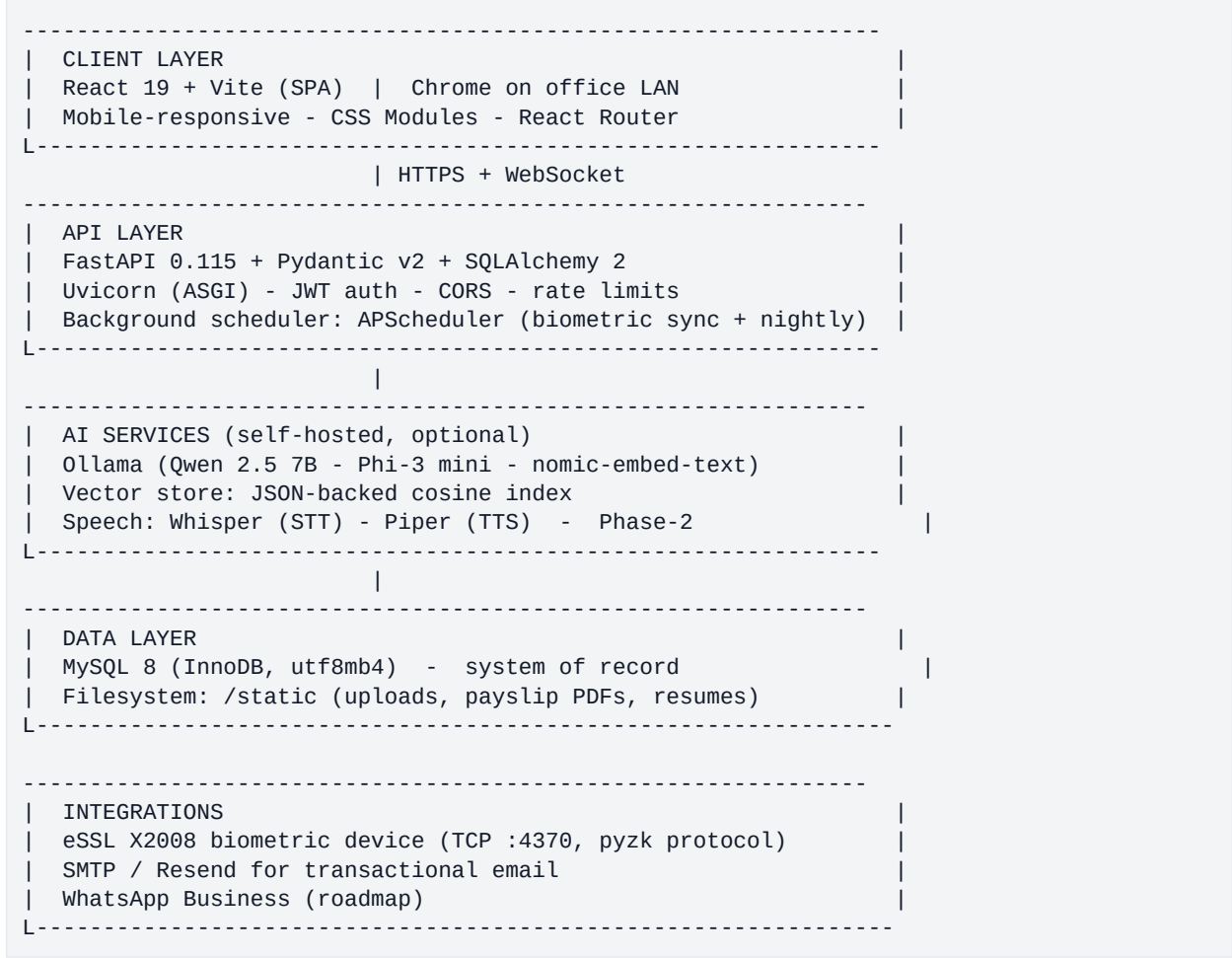
3. Guiding Principles

The system was designed against the following non-negotiable principles:

- **On-prem first** - payroll, salary, appraisal, and health data must not leave the office. No third-party AI APIs, no cloud SaaS dependencies.
- **Grounded AI, not hallucinated** - every factual response the AI produces must cite a source document or database row.
- **Role-aware from the first token** - an employee can never see peers' salaries; a manager cannot approve their own leave.
- **Observable end-to-end** - every mutation is auditable and reversible.
- **Terse, professional UI** - no emojis in the product surface; inline SVG icons only; BVC red + white + gold palette.
- **Additive over destructive** - new features never break existing modules; auto-migrations handle schema evolution.

4. System Architecture

BVC24 ERP is a **three-tier web application** with an optional AI services tier:



5. Technology Stack

The stack was selected for zero-license-cost, on-prem operability, and long-term community support.

Layer	Technology	Rationale
Frontend	React 19 + Vite + React Router 7	Fast HMR, small bundle, mature ecosystem
UI styling	CSS Modules (no framework)	Scoped styles, no runtime cost, no vendor lock-in
Backend	FastAPI 0.115 + Pydantic v2	Type-safe request/response, OpenAPI auto-generated
ORM	SQLAlchemy 2.0	Battle-tested, expressive query API
Database	MySQL 8.0 (InnoDB)	Familiar SQL surface, robust replication
Auth	JWT (python-jose) + bcrypt	Stateless tokens, industry standard
PDF generation	ReportLab + markdown	Pure-Python, no external binaries
Biometric bridge	pyzk 0.9 (Python)	Open-source ZKTeco / eSSL protocol client
Background jobs	APScheduler 3.10	In-process, no separate worker to manage
LLM runtime	Ollama (Qwen 2.5 7B, Phi-3 mini)	Local inference, no API keys
Embeddings	nomic-embed-text	Multilingual, permissive licence
Deployment	Ubuntu 22.04 - Jenkins - systemd - Nginx	Standard on-prem Linux stack
CI/CD	Jenkins pipeline (Git -> build -> deploy)	Already deployed by BVC IT

6. Module Catalogue

The system is organised into six functional families, mirroring the sidebar navigation.

6.1 HRMS - Human Resource Management

Module	Purpose
RBAC / Role Management	Fine-grained permissions per role. Every route and every AI tool call passes through a permission check.
Employees	Complete personnel master - profile, documents, salary structure, RBAC role.
Memos	Internal broadcast messages with priority, department scope, and read acknowledgement.
Attendance	Live biometric-driven attendance with LATE tracking (9:15 grace), overtime, and geofence fallback.
Shift Management	Shift definitions and per-employee shift assignments.
Leave Management	Casual / Sick / Earned leave with balances, approval workflow, and email notifications.
Payroll	Salary components (BASIC + HRA + DA + allowances), PT / PF / ESI configuration.
Star Performance	4-dimension scoring (Attendance / Task / Leave / Permission) with monthly stars, prorated for the current month, nightly auto-recompute.
Allowances	Custom recurring allowances per employee.
Recruitment	Job postings, candidate pipeline, application tracking.
Onboarding	Token-based invite flow - HR emails a link; candidate self-registers with password + profile.
HR Automation	AI-assisted memo generation, complaint routing, offer letter drafting.
Monthly Reports	Auto-generated per-employee monthly summary - working days, present, absent, leave breakdown, deductions, net payable. Downloadable as PDF.

6.2 CRM & Sales

Module	Purpose
Customer master	Full contact history, GST, addresses, credit limit
Lead pipeline	New -> Contacted -> Qualified -> Won / Lost with follow-up dates
Quotations	Line-item quotes with public share-link and email tracking
Sales Orders	Confirmed orders with advance-payment tracking
Invoices	Tax-compliant invoice generation

6.3 Project & Manufacturing

Module	Purpose
Projects	Kanban board, task assignments, star statuses, deletion controls
Product Models	Machine catalogue (Snack Combo, Beverage, Medical Dispenser, etc.)
BOM (Bill of Materials)	Parts and quantities per model
Work Orders	Manufacturing runs tied to sales orders
Process Stages	Per-work-order production stages with assignee and completion tracking

6.4 Purchase & Inventory

Module	Purpose
Suppliers	Vendor master, payment terms, past PO history
Purchase Orders	Draft -> Approved -> Received lifecycle
GRN (Goods Received Notes)	Inspection and stock-in logging
Inventory	Live stock levels with reorder-point alerts

6.5 Reports & Analytics

Module	Purpose
Dashboards	Enterprise Command Center - KPIs across HR, sales, manufacturing
Attendance reports	Daily, monthly, per-employee
Payroll reports	Monthly attendance report tied to salary calculation
Sales analytics	Revenue trends, lead conversion, top customers

6.6 System

Module	Purpose
Company Settings	Master company profile, GST, addresses, logos
Holiday Calendar	Vendor-scoped declared holidays used by attendance and payroll calculations
Geofence Settings	Office coordinates + radius for GPS-gated attendance
Settings	Office hours, late grace, permission cap, etc.

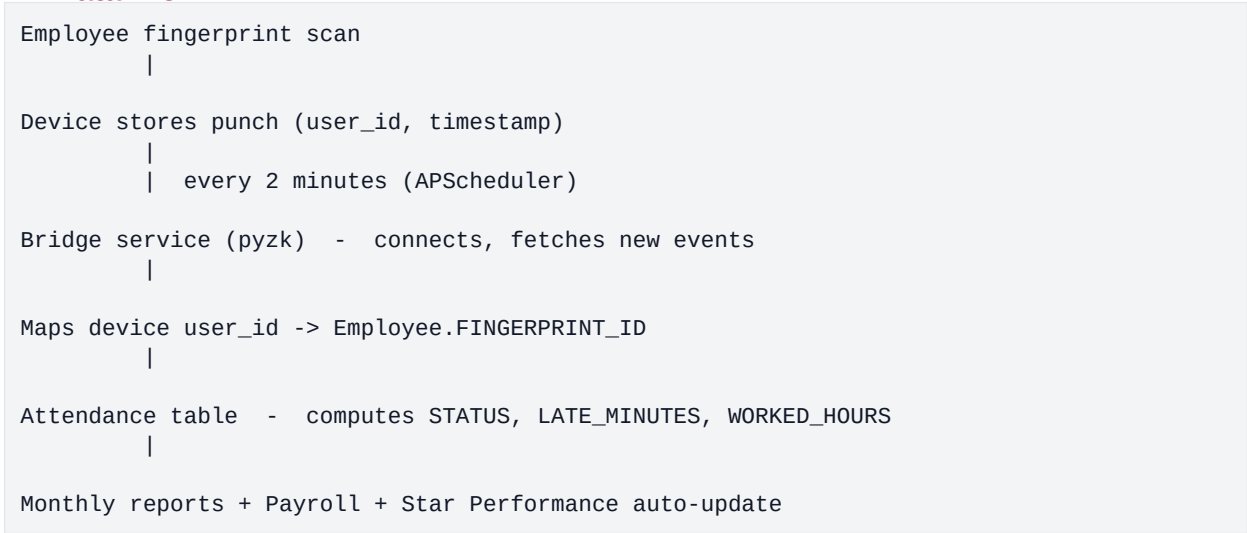
7. Biometric Attendance Integration

The system integrates directly with an **eSSL X2008** biometric fingerprint device deployed at the office entrance.

7.1 Device Specifications

Property	Value
Model	eSSL X2008
Firmware	Ver 8.0.4.7 (2025-02-12)
Fingerprint algorithm	VX10.0
Serial number	JNP2255102739
Communication	TCP/IP on port 4370
Office LAN address	192.168.1.201

7.2 Data Flow



7.3 Attendance Rules

- **Shift hours:** 9:00 AM - 6:00 PM
- **Late cutoff:** 9:15 AM (15-minute grace period; anyone punching after 9:15 is marked LATE with minute-count stored)
- **Overtime:** any work past 6:00 PM triggers OT tracking on OT check-in / OT check-out punches
- **Sundays:** excluded from working days; Sunday punches are logged but don't contribute to attendance count
- **Holidays:** configured in Holiday Calendar; deducted from monthly working days

7.4 Automatic Sync

- Every 2 minutes while the backend is running, the bridge polls the device for new punches
- Idempotent - running twice on the same events produces zero duplicates
- Watermark-based - resumes from the last successful event after a restart
- Read-only - never mutates or clears device storage; the device remains the source of truth

8. AI-Powered HR Assistant

BVC24 ERP includes a self-hosted AI assistant for HR-related queries. Every AI feature runs on the office server - **no data ever leaves your infrastructure**.

8.1 Current Capabilities (Phase 1)

- **Employee chatbot** - natural-language Q&A over HR policy documents (RAG)
- **Leave query bot** - "how many casual leaves do I have left?", "when did I last take sick leave?"
- **Payroll query bot** - "explain last month's salary breakdown"
- **HR policy assistant** - cites the policy document and paragraph for every answer
- **Resume parser** - extracts structured data from PDF resumes (name, skills, experience, education) using Qwen 2.5's JSON mode
- **Onboarding AI service** - assists candidates through the self-registration flow

8.2 Technology Choices

Component	Choice	Notes
LLM (chat)	Qwen 2.5 7B Instruct via Ollama	Multilingual, JSON-mode, ~4.5 GB Q4 quantised
LLM (fast fallback)	Phi-3 mini 3.8B via Ollama	Sub-second on CPU
Embeddings	nomic-embed-text	Multilingual dense embeddings
Vector store	JSON-backed cosine index	Zero infra - file on disk
Corpus source	Policy PDFs + DB tables	Auto-indexed on startup

8.3 Voice-First Roadmap (Phases 2 - 6)

The Software Requirements Specification for the full 30-feature AI HR Assistant lives at `docs/AI_HR_ASSISTANT_SRS.md`. Highlights:

- **Voice-first, text-fallback** interaction - Whisper (STT) + Piper (TTS)
- **Multilingual** - English + Tamil + Hindi with automatic language detection
- **Function-calling agents** - LLM emits structured tool calls (`book_leave`, `get_payslip`, `apply_permission`) with RBAC guards
- **Predictive ML** - attrition risk, attendance anomaly, leave prediction, payroll anomaly (LightGBM / XGBoost)
- **HR Copilot sidebar** - inline "what should I do next?" for managers
- **Speaker verification** - voice-based employee authentication (ECAPA-TDNN)

Full 6-phase roadmap detailed in the SRS document.

9. Database Design

BVC24 ERP uses a normalised relational schema with 60+ tables, all under a single MySQL database `vending_erp`. Every business table carries a `VENDOR_ID` column for multi-tenant scoping.

9.1 Core Table Groups

- **Identity & auth:** `employee`, `role`, `permission`, `role_permission`, `department`, `designation`
- **Attendance:** `attendance`, `attendance_security_log`, `biometric_event`, `geofence_settings`, `holiday_calendar`

- **Leave:** leave_request, leave_balance, leave_quota_policy
- **Payroll:** salary_structure, payroll_run, salary_component, payslip
- **Performance:** monthly_attendance_report, performance_score, attendance_alert
- **Task & Project:** project, task_assignment, daily_allocation, process_stage, work_order, work_order_stage_progress
- **Manufacturing:** product_model, bom_item, machine
- **Sales:** customer, quotation, sales_order, invoice
- **Purchase:** supplier, purchase_order, grn, supplier_payment
- **Onboarding:** employee_onboarding_session, onboarding_checklist_item, employee_document
- **Comms:** notification, memo, employee_memo
- **AI:** ai_conversation, ai_turn, ai_tool_call, voice_enrollment, ml_model_registry

9.2 Referential Integrity

- Foreign keys enforced at the DB layer with ON DELETE RESTRICT on most business links
- Auto-migration on backend startup: scans INFORMATION_SCHEMA and adds missing columns / drops stale indexes / rebinds broken foreign keys
- Every writable table has CREATED_AT and UPDATED_AT audit columns

9.3 Backup Strategy

- Nightly mysqldump (recommended) archived to a network share
- Weekly full-DB snapshot to an external drive
- Point-in-time recovery via MySQL binary log

10. API Reference - Selected Endpoints

The complete OpenAPI schema is served at <http://<backend>:8001/docs> (Swagger UI). Highlights:

10.1 Authentication

Method	Path	Purpose
POST	/admin-login	Admin / HR credentials -> JWT
POST	/employee-login	Employee credentials -> JWT
POST	/employee-logout	Marks logout time

10.2 Attendance

Method	Path	Purpose
POST	/check-in	Geofenced manual check-in (fallback when biometric unavailable)
POST	/check-out	Geofenced manual check-out
GET	/attendance	Filterable history (admin)
GET	/attendance/today	Today's rows (self for employees, all for admins)
GET	/attendance/my-history	Own-history (employees)
POST	/biometric/scan	Device webhook (normalised payload)

10.3 Leave & Permissions

Method	Path	Purpose
POST	/leave/apply	Submit leave request
POST	/leave/apply-permission	Submit hourly permission (4h monthly cap)
GET	/leave/balance/{employee_id}	Current leave balance
GET	/leave/permission-balance/{employee_id}	Current permission usage
PATCH	/leave/{id}/approve	Manager approves

10.4 Reports

Method	Path	Purpose
GET	/monthly-reports?year=&month=	Monthly attendance + payroll
POST	/monthly-reports/generate	Force recompute for a month
GET	/monthly-reports/{emp_id}/pdf	Downloadable PDF
GET	/performance/stars?year=&month=	Star performance scores

10.5 AI

Method	Path	Purpose
POST	/ai-chat	Natural language query
POST	/leave-brain/*	Leave-specific reasoning

Every endpoint is protected by JWT + RBAC. Denied requests return 403 with an audit trail entry.

11. Security & Compliance

11.1 Authentication & Authorisation

- Short-lived JWTs (default 8-hour expiry), rotated via refresh tokens
- Passwords hashed with **bcrypt** (cost factor 12)
- Role-based access control on every route: `Depends(require("permission.name"))`
- SSO-ready hook for SAML / OIDC (roadmap)

11.2 Data Protection

Concern	Control
Passwords	bcrypt-hashed, never returned in API responses
PII (Aadhaar, PAN, phone)	Masked in audit logs before persistence
Salary data	Only visible to HR_ADMIN and above; per-employee scope for others
Payslip PDFs	Served via authenticated endpoint, never publicly linkable
Onboarding invite tokens	UUID-based, single-use, 7-day expiry

11.3 Audit Trail

- Every mutation logged to `attendance_security_log` or the module-specific audit table
- AI conversations recorded in `ai_turn` + `ai_tool_call` with input/output/latency
- Append-only design - no UPDATE or DELETE on audit tables

11.4 On-Premise Guarantee

- Zero calls to external AI APIs - Ollama and all model weights are local
- No cloud storage - files served from local filesystem via authenticated endpoint
- Air-gap ready - the system runs without internet access (only SMTP for email is external)

12. Deployment

12.1 Target Environment

- **Server:** Ubuntu 22.04 LTS (recommended) or Windows Server 2019+
- **Hardware:** 8-core CPU, 32 GB RAM, 500 GB SSD, optional NVIDIA GPU (12 GB VRAM) for larger LLMs
- **Network:** office LAN with static IP for the server (typically `192.168.1.10`)
- **Concurrent capacity:** 50 simultaneous voice sessions or 500 text queries per minute on the recommended hardware

12.2 Component Layout

```
Ubuntu server (192.168.1.10)
|--- MySQL 8          -> :3306 (localhost only)
|--- FastAPI (uvicorn) via systemd -> :8001 (behind Nginx)
|--- Ollama           -> :11434 (localhost only)
|--- Nginx            -> :80 / :443 (public LAN)
|--- /var/www/bvc-erp -> React build served by Nginx
L-- /opt/bvc-erp      -> application code, .env, virtualenv
```

12.3 CI/CD Pipeline (Jenkins)

The Jenkinsfile at the repo root defines the pipeline:

```
Stage 1: Checkout      -> git clone
Stage 2: Backend        -> pip install -r requirements.txt
Stage 3: Frontend       -> npm ci && npm run build
Stage 4: Deploy         -> rsync to /opt/bvc-erp + /var/www/bvc-erp
Stage 5: Restart        -> systemctl restart bvc-backend
```

Every push to `main` triggers a pipeline run. Duration: ~90 seconds.

12.4 Environment Variables (`.env`)

Managed via Jenkins credentials store; never committed to Git. Key settings:

```
MY_SQL=mysql+pymysql://erpdbuser:PASSWORD@localhost
DB_NAME=vending_erp
JWT_SECRET_KEY=<long-random>
SMTP_HOST=smtp.gmail.com
SMTP_USER=support@bvc24.com
RESEND_API_KEY=<optional>
ESSL_DEVICE_IP=192.168.1.201
ESSL_COMM_KEY=123456
```

13. Automated Background Jobs

APScheduler runs three critical background tasks inside the FastAPI process:

Job	Schedule	Purpose
essl-autosync	Every 2 minutes	Pull new biometric punches from the eSSL X2008 device
star-autosync (boot)	30 s after every backend startup	Recompute current month's Star Performance scores
star-autosync (nightly)	Daily 01:15 IST	Recompute Star Performance using yesterday's finalised attendance

No manual triggering required. All jobs are idempotent and safely resume after a server restart.

14. User Roles

The system ships with the following out-of-the-box roles. Additional roles can be created via the RBAC page.

Role	Access Scope
SUPER_ADMIN	Everything, including RBAC editing

Role	Access Scope
HR_ADMIN	Full HR access + read-only elsewhere
MANAGER	Team-level HR + task management
EMPLOYEE	Own data only - attendance, leave, payslip, tasks
SALES_EXEC	CRM + own quotations
PROD_SUPERVISOR	Manufacturing + task allocation
ADMIN	System settings + read-only elsewhere

15. Employee Self-Service Portal

Every employee lands on the Welcome page after login. Six primary cards (all clickable, no sidebar):

1. **Attendance** - today's punches, working hours, monthly history
2. **Tasks** - assigned tasks with Start / Hold / Done actions
3. **Leave** - apply, track, cancel; balance shown
4. **Permission** - hourly time-off requests (4h monthly cap)
5. **Memo** - company memos and notices
6. **Star Performance** - personal monthly score and dimension breakdown
7. **Payslip** - downloadable monthly PDF

All flows are mobile-responsive and available on the office WiFi via phone.

16. Manager Dashboards

Managers see additional widgets on the Enterprise Command Center:

- **Team attendance snapshot** - who's in, who's late, who's absent
- **Pending approvals** - leave requests, task acceptances, expense claims
- **Team performance** - Star Performance rollup across direct reports
- **Task Kanban** - drag-and-drop board of team tasks

17. Admin Runbook

17.1 Daily Operations

- Backend unicorn service runs continuously via `systemd`. Check status: `systemctl status bvc-backend`
- Biometric sync runs automatically every 2 minutes
- Star Performance recomputes at 01:15 daily
- No manual intervention required for routine days

17.2 Common Tasks

Task	Command
View live backend logs	<code>journalctl -u bvc-backend -f</code>
Restart backend	<code>sudo systemctl restart bvc-backend</code>
Run one-off biometric sync	<code>python -m app.services.essl_bridge</code>
Bulk import employees from device	<code>python -m scripts.seed_from_essl --apply</code>
Recompute Star Performance	Auto - via daily job; manual: <code>python -m scripts.recompute_stars</code>
Backup MySQL	<code>mysqldump vending_erp \ gzip > backup.sql.gz</code>

17.3 Troubleshooting Playbook

Symptom	Investigation
Backend won't start	<code>journalctl -u bvc-backend -n 100</code> - read the traceback
Attendance not syncing	ping the device; check <code>essl_bridge</code> logs for network errors
Employees showing ■0 salary	Confirm <code>salary_structure</code> row exists; check the Payroll module
Monthly report row stuck	Un-lock via SQL: <code>UPDATE monthly_attendance_report SET STATUS='GENERATED' WHERE ...</code>
Star Performance empty	Confirm attendance rows exist for the period; wait for next tick or hit the page

18. Testing & Quality

- **Backend tests:** pytest suite covering RBAC, auth, attendance rules, leave calculations, biometric mapping
- **Frontend build:** Vite production build with tree-shaking, minification, and asset hashing
- **Auto-migrations:** verified on startup - no manual schema changes needed after code updates

19. Roadmap

19.1 Immediate (Q3 2026)

- Full voice-first AI HR Assistant (Phases 1-3 of the SRS)
- Predictive ML models for attrition and attendance anomaly
- HR analytics dashboard with natural-language query interface

19.2 Medium-term (Q4 2026 - Q1 2027)

- Meeting assistant (record -> transcribe -> summarise -> action items)

- AI career growth advisor with skills graph
- Multi-device biometric support (multiple eSSL / ZKTeco readers)
- Mobile PWA with offline attendance queueing

19.3 Long-term

- Multi-tenant SaaS deployment (already scoped in the schema via `VENDOR_ID`)
- SSO integration (SAML / OIDC)
- WhatsApp bot for employee self-service
- Air-gapped deployment package for compliance-sensitive customers

20. Support & Contact

- **Product owner:** Bharath Vending Corporation
- **Development contact:** BVC24 IT Team
- **Repository:** internal Git (private)
- **Bug reports:** raise via Jenkins issue tracker or email
- **User training:** on-site quarterly refresher sessions

Appendix A - Glossary

Term	Meaning
RBAC	Role-Based Access Control
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
STT / TTS	Speech-to-Text / Text-to-Speech
ORM	Object-Relational Mapper
BOM	Bill of Materials
GRN	Goods Received Note
ESI / PF / PT	Employee State Insurance / Provident Fund / Professional Tax
SRS	Software Requirements Specification
ADMS	Auto Data Management Service (eSSL push protocol)

Appendix B - File Structure

```
Vendor-based Manufacturing ERP/
|--- backend/
|   |--- app/
|   |   |--- main.py           # FastAPI entry + auto-migrations
|   |   |--- database/        # DB connection
|   |   |--- models/          # SQLAlchemy models (60+ tables)
|   |   |--- routes/          # HTTP endpoints (grouped by module)
|   |   |--- services/        # Business logic
|   |   L-- schemas/          # Pydantic request/response schemas
|   |--- scripts/             # Migration + one-off tools
|   |--- venv/                 # Python virtual environment
|   |--- requirements.txt
|   L-- .env                   # Secrets - never committed
|--- frontend/
|   |--- src/
|   |   |--- pages/            # Route components
|   |   |--- components/       # Reusable UI
|   |   |--- services/api.js   # Axios client
|   |   |--- utils/            # Shared helpers
|   |   L-- App.jsx
|   |--- public/
|   |--- package.json
|   L-- vite.config.js
|--- docs/                    # This file lives here
|--- Jenkinsfile               # CI/CD pipeline
L-- README.md
```

End of document. Prepared for Bharath Vending Corporation, 10 July 2026.